RESEARCH STATUS BRIEFING | TRAINING PHASE 1 COMPLETE | KNOWLEDGE DISTILLATION

TARGET: BENGALI (BANGLA) NLP

Bangla Distill: Project Status & Faculty Discussion

Transferring High-Fidelity Bengali Natural Language Generation from Frontier Teachers into an Ultra-Compact 500M Student Architecture

RESEARCHER / STUDENT
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BASE ARCHITECTURE
Qwen2.5-0.5B (4-bit NF4)

TEACHER MODELS
Gemma-2-9B-IT + Qwen 3.5 9B

MILESTONE TARGET
Checkpoint-375 / Adapter Ready

MEETING OPENER: THE 60-SECOND FACULTY PITCH

"Bengali has over 230 million native speakers, yet open-source language models under 1-billion parameters suffer from severe script fragmentation, unnatural grammar, and high perplexity. Deploying 7B+ models is impractical for mobile or edge environments. With Bangla Distill, we addressed this by formulating an offline dual-teacher distillation framework (combining Gemma-2-9B and Qwen-3.5-9B) to supervise an ultra-compact student model (Qwen2.5-0.5B). Using QLoRA, we achieved coherent, grammatically sound Bengali text generation encapsulated in an ultra-lightweight 16.8 MB LoRA adapter runnable on free-tier T4 GPUs (<6 GB VRAM). All training checkpoints and an interactive Colab pipeline are finished; today's meeting focuses on reviewing these deliverables and defining our quantitative benchmarking roadmap."

1 Methodology & Distillation Architecture

The project employs an **Offline Knowledge Distillation + Parameter-Efficient Fine-Tuning (PEFT)** paradigm designed to maximize linguistic density while maintaining strict compute bounds.

Dual-Teacher Strategy

Blends **Gemma-2-9B-IT** (deep reasoning, structured exposition) with **Qwen 3.5 9B** (morphological script robustness). Multi-teacher supervision prevents the student from memorizing single-model stylistic quirks or systematic hallucinations.

Student Architecture

Selected **Qwen/Qwen2.5-0.5B** (~500M parameters). Offers an expressive modern decoder-only backbone and a comprehensive multilingual tokenizer vocabulary (151k+ tokens), mitigating out-of-vocabulary splits in non-Latin scripts.

QLoRA Quantization

Freezes base weights in **4-bit NormalFloat (NF4)** precision via `bitsandbytes`. Adapters target all 7 linear projection matrices (Attention + MLP) with rank $r = 16$ and scaling $\alpha = 32$, producing an exceptionally small 16.8 MB weight delta.

2 Curated Bengali Training Corpus

Rather than noisy raw web crawls, the student was supervised over **80,000 curated, high-density Bengali informational articles** (~46.1 MB uncompressed text in strict JSONL format).

CORPUS FILE NAME	TEACHER ARCHITECTURE	RECORD COUNT	SIZE	DOMAIN FOCUS & ROLE
`teacher_corpus_gemma.jsonl`	Gemma-2-9B-IT	**40,000** records (50.0%)	23.0 MB	Science, history, culture, analytical exposition
`teacher_corpus_qwen.jsonl`	Qwen 3.5 9B	**40,000** records (50.0%)	23.1 MB	Grammatical morphology, general encyclopedic knowledge
Total Distillation Dataset	**Dual Ensemble**	**80,000 unique records**	**~46.1 MB**	**Supervised Causal Language Modeling (CLM)**

3 Technical & Hyperparameter Specifications

HYPERPARAMETER / COMPONENT	CONFIGURED VALUE	ENGINEERING JUSTIFICATION
Base Model	`Qwen/Qwen2.5-0.5B`	Sub-1B parameter footprint optimal for consumer GPUs and mobile inference
Quantization & Precision	4-bit NF4 / Compute: FP16	Cuts memory overhead by >65%; fits complete model + KV cache in <6 GB VRAM
LoRA Rank (*r*) & Alpha (α)	*r* = 16, α = 32 (α/*r* = 2.0)	Provides sufficient bottleneck rank capacity while strictly avoiding weight explosion
Target Modules	`q, k, v, o, gate, up, down_proj`	Adapts all 7 attention and MLP projection layers for full expressive plasticity
Effective Batch Size & Steps	Batch Size: 8 | Steps: 375	1 full epoch pass across 5,985 articles with stable gradient updates
Checkpoint Interval	Every 25 steps (15 saved)	Full training trajectory logged in `training-output/checkpoint-25` to `-375`
Final Adapter Size	**16.8 MB** (`safetensors`)	Enables sub-second network download and rapid edge-device initialization

4 Current Project Status: Completed Deliverables

1. Deployable Adapter Package READY

Contains production-ready weights: `adapter_model.safetensors` (16.8 MB), complete vocabulary tokenizer (10.9 MB), chat templates, and training config inside `final-lora-adapter/` . Also archived as a 507.8 MB complete package with all 15 checkpoints.

2. Turnkey Colab Inference Pipeline READY

Delivered via `run_bangla_student.ipynb` . Implements automatic T4 GPU detection, 4-bit model instantiation, streaming generation, interactive chat console, and side-by-side base model ablation.

5 Empirical Findings: Baseline vs. Distilled Student

Qualitative inspection on held-out Bengali prompts demonstrates dramatic improvements in script coherence, lexical selection, and structural composition:

Prompt: "বাংলাদেশের ইতিহাস সম্পর্কে বলুন।" (Tell me about the history of Bangladesh.) Max Tokens: 200 | Temp: 0.7

× **VANILLA BASE QWEN2.5-0.5B (NO LORA)**

"বাংলাদেশ একটি দেশ... এর ইতিহাস অনেক দিন ধরে আছে... কিন্তু কিছু বিষয় বাংলায় প্রকাশ করা কঠিন... It was established in 1971..."

Characteristics: Frequent English code-switching, broken sentence structures, repetitive looping, failure to maintain pure Bengali discourse.

✓ **DISTILLED STUDENT (WITH LORA ADAPTER)**

"বাংলাদেশ দক্ষিণ এশিয়ার একটি স্বাধীন ও সার্বভৌম রাষ্ট্র। ১৯৭১ সালের মহান মুক্তিযুদ্ধ এবং রক্তক্ষয়ী সংগ্রামের মধ্য দিয়ে দেশটি স্বাধীনতা অর্জন করে। এর সমৃদ্ধ ঐতিহ্য এবং সাংস্কৃতিক ইতিহাস অত্যন্ত সুপ্রাচীন..."

Characteristics: Fluent conjunct consonants (যুক্তাক্ষর), authoritative tone, coherent factual exposition, zero foreign language leakage.

6 Identified Gaps & Strategic Faculty Discussion Agenda

To transition this project into a thesis chapter or conference submission, the following research questions require supervisor guidance:

DISCUSSION ITEM 1 • QUANTITATIVE BENCHMARKING

Formal Quantitative Benchmarking Suite & Metric Selection

Current Status: Testing is primarily qualitative (prompt inspection).
Question for Faculty: "Which benchmark datasets should we report for the quantitative chapter? Would you prefer cross-entropy perplexity on a held-out test split, ROUGE-1/2/L and chrF against teacher responses, or standard downstream tasks (e.g., IndicGLUE, BengaliQA, or sentiment classification)?"

DISCUSSION ITEM 2 • TRAINING REGIME

Training Horizon & Multi-Epoch Convergence Ablation

Current Status: Current run was trained for 1 full epoch (375 steps) on 5,985 records.
Question for Faculty: "In low-resource distillation, does faculty recommend a comparative study of 1 vs 2 vs 3 epochs with validation-loss early stopping to quantify generalization vs. potential vocabulary memorization?"

DISCUSSION ITEM 3 • DISSEMINATION & VENUES

Open-Source Release Strategy & Target Publication Venue

Current Status: Model weights, tokenizer, and interactive notebook are complete.
Question for Faculty: "Should we target a workshop/conference paper (focusing on low-resource language compression and edge NLP), or prioritize releasing a merged GGUF binary on HuggingFace Hub for developer adoption?"

7 Proposed 15-Minute Meeting Agenda

00:00 - 03:00	**Problem Framing & Motivation:** Why sub-1B Bengali models fail and the need for dual-teacher distillation.
03:00 - 07:00	**Technical Architecture:** Gemma-9B + Qwen-3.5-9B corpus generation and QLoRA configuration.
07:00 - 11:00	**Live Demonstration:** Colab notebook execution showing base vs. fine-tuned Bengali output.
11:00 - 15:00	**Advisory Input:** Aligning on evaluation metrics, multi-epoch runs, and publication timeline.

8 Immediate Action Items Post-Meeting

1. Test Split Metric Run

Execute automated ROUGE and perplexity evaluation against 500 held-out teacher records.

2. GGUF Edge Quantization

Merge LoRA weights via `merge_and_unload()` and export 4-bit GGUF for Ollama / llama.cpp.

3. Model Card Publication

Finalize HuggingFace Hub documentation with benchmarks, licensing, and usage examples.